

Next-Generation Systems

High-Performance Scientific Presentations & Typesetting

Presenter: Systems Engineering Research Lab

Date: \today

Typography & Text Formatting

Botox provides full Markdown and LaTeX styling for presentations:

- **Emphasis:** Use **bold**, *italics*, or ***bold-italics*** for key terms.
- **Formulas & Chemistry:** Water H_2O , energy $E = mc^2$, and powers 2^{10} .
- **Revisions:** Flag deprecated items with ~~strikethrough~~.
- **Commands:** Run inline code with `botox slides.md -o slides.pdf`.
- **Workflow:** - Initialize workspace with `botox init --slides`
 - Live preview with synchronized bidirectional cursor tracking

Callout Boxes & Admonitions

Visual callouts highlight critical insights and best practices:

“[!NOTE] Botox compiles Markdown to vector Typst slides in milliseconds with zero external toolchain dependencies.

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“[!TIP] Switch themes instantly between `default`, `academic`, `nord`, and `dark` via the frontmatter `theme` property.

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Mathematical Formulations

Native equations render with Computer Modern vector precision:

Scaled Dot-Product Attention maps queries Q and keys K to values V :

$$\text{Attn}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

- Dimension scaling $\frac{1}{\sqrt{d_k}}$ avoids vanishing gradients in large models
- Fully vector-rendered with zero pixelation at any display scale

Matrix Models & Linear Algebra

High-dimensional state representations and parameter covariance:

$$\mathbf{\Sigma} = \begin{bmatrix} \sigma_x^2 & \rho_{xy} & 0 \\ \rho_{yx} & \sigma_y^2 & 0 \\ 0 & 0 & \sigma_z^2 \end{bmatrix}, \quad \nabla_{\boldsymbol{\theta}} \mathcal{L}(\boldsymbol{\theta}) \in \mathbb{R}^d$$

- Seamless LaTeX matrix syntax with `\begin{bmatrix} ... \end{bmatrix}`
- Complete support for Greek symbols, subscripts, superscripts, and norms

Code Blocks & Syntax Highlighting

Vector monospace rendering with automated contrast and framing:

```
use botox::compiler::compile_typst;

// High-speed slide compilation directly to vector PDF
pub fn build_deck(src: &str, out: &std::path::Path) -> Result<(), String>
{
    compile_typst(src, out, None)
}
```

Quantitative Performance

Compilation speed and resource efficiency comparisons:

Pipeline Engine	Compilation	Memory	Output Format	Math Quality
Botox (Native)	18 ms	~24 MB	PDF & SVG	Exact Vector
Pandoc + LaTeX	1,850 ms	~320 MB	PDF only	Native TeX
Marp CLI + Chrome	2,400 ms	~450 MB	HTML / PDF	Rasterized
Quarto + Typst	420 ms	~110 MB	PDF / HTML	Native Typst

Interactive Citations & References

Hyperlinks automatically transform into IEEE-formatted citations:

- Scalable transformer models introduced by Vaswani et al. [1].
- Deep residual learning architectures evaluated in He et al. [2].
- Sparse mixture-of-experts routing detailed in Fedus et al. [3].
- Native layout compiler design inspired by Typst Project [4].

All citation links remain clickable in exported PDF and SVG presentations.

Summary & Deployment

Key Takeaways

- **Unified Format:** Share syntax between academic papers and slide decks.
- **Sub-Second Feedback:** Instant live reload during writing and editing.
- **Publish Anywhere:** One-command export to PDF, SVG, and interactive preview.

“[!IMPORTANT] Try other built-in slide themes by changing `theme: nord` or `theme: dark` in the frontmatter!

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References

- [1] “Vaswani et al.”, [Online]. Available: <https://arxiv.org/abs/1706.03762>.
- [2] “He et al.”, [Online]. Available: <https://arxiv.org/abs/1512.03385>.
- [3] “Fedus et al.”, [Online]. Available: <https://arxiv.org/abs/2101.03961>.
- [4] “Typst Project”, [Online]. Available: <https://typst.app>.